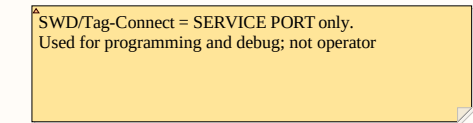


Essential performance (controller role):

- Board provides BLE command/control only; no direct patient-applied parts.
- All MCU I/Os are SELV and default to safe states on reset (pull-ups/-downs and series resistors as shown).
- BLE link integrity and stimulation safety are handled at system/implant level.

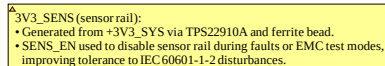
Design follow AN5165 rf hardware STM32WB.pdf

Diagram illustrating the RF hardware for STM32WB, showing the connection of the DLF162500LT-5028A1 component to the RF1_FLT and RF1_ANT pins. The component is connected to ground (GND_1, GND_2) and the RF1_FLT pin. The RF1_ANT pin is connected to the ANT1 component (2450AT18A100E) and the ESD_RF1 component (PESD5V0S1UL,315).



- BOOT0 (PH3) — Boot strap
- 100 k Ω to GND (default: User Flash).
- TP to +3V3 to force System Bootloader.
- Do not use as GPIO.

Sensor Power Gate



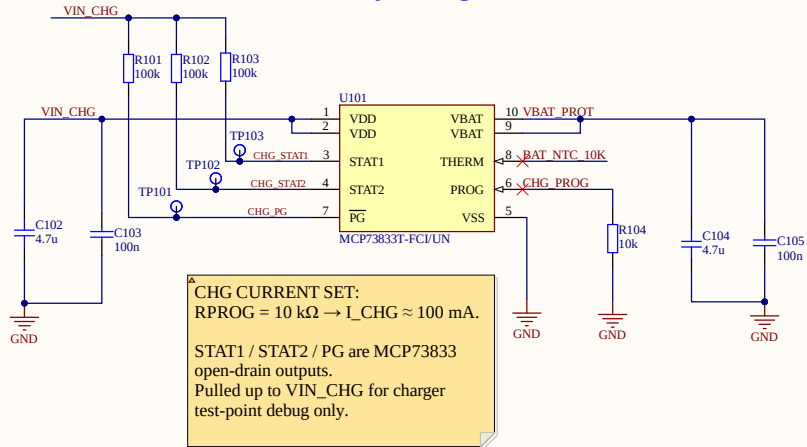
- Operator I/O (button + LED + skin sensor):
 - Button: TVS at pad + 100R series + RC network for ESD and burst immunity.
 - Sensors: on-board only, short I2C, powered via 3V3_SENS gate.
 - No direct patient-applied parts on this PCB; temperature is read through enclosure/assembly.
- BTn1 default state:
 - Pull network ensures defined logic level on reset and under noise.



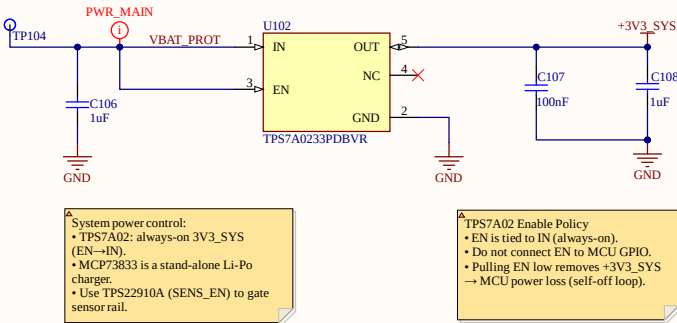
Title BLE-Control Wearable — Schematic Set (AD25)			
Size A3	Number BLEC-SCH-0003	Revision A0 (EVT)	
Date: 5/26/2026	Sheet of 3 of 3		Drawn By: Caitie Donohoe
File: C:\Users\LED_Sch\Documents\BLEC-SCH-0003\BLEC-SCH-0003_Schematic_Set.dwg			

Power, Battery, USB-C & MCP73833T (Main 3V3) — BLEC-SCH-0001 Non-medical wearable prototype

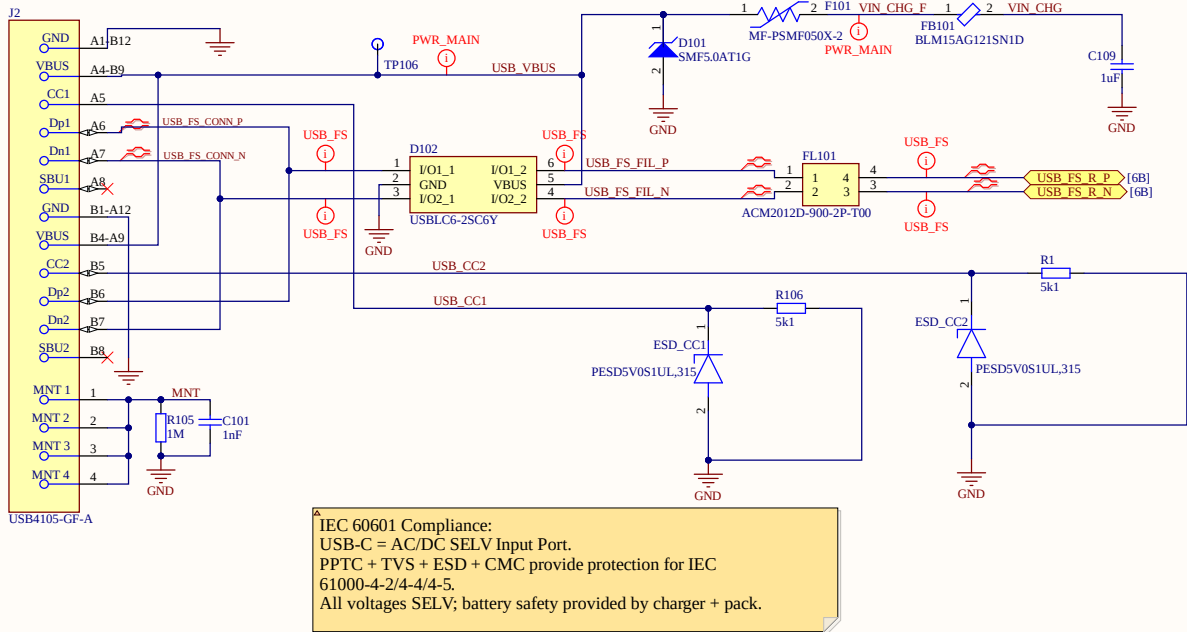
Battery Charge Cct



REG LINEAR



USB Filtering and ESD



Battery Connect

